

# SAM BAI

## PRODUCT ENGINEER

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### EXPERIENCE

#### Founder & Product Engineer | Solynt Labs Limited

2026 - Present

Founded and run a New Zealand software studio; lead product discovery, architecture, implementation, deployment and client support.

#### RIVET | Construction operations PWA - client delivery

Jul 2026 - Present

- Built and deployed a private construction-operations PWA for a New Zealand residential builder, joining project, task, exception, approval, calendar and notification workflows for directors and project managers.
- Designed server-authoritative schedule proposals that recalculate before apply, reject stale changes and write dates, audit records and notification effects atomically in PostgreSQL.
- Implemented scoped authentication and conflict-aware browser sync with IndexedDB outboxes, version checks and transactional Hono APIs on Cloudflare Workers.

#### TREKKY | Job-search workspace - owned product

Jan 2026 - Present

- Built and operate a live Next.js/TypeScript product for job discovery, application tracking, contacts, AI preparation, outcome analytics and review-first automation.
- Added account-scoped MCP access, a Chrome MV3 capture extension and multi-source ingestion across NZ, Australia, the US and Singapore with canonical deduplication and provenance.
- Cut default dashboard JavaScript from 341 KB to 233 KB gzip through route-level lazy loading.

### SELECTED ENGINEERING

#### CLEARFOLD | Pre-production digital-asset operations

2026

- Built exact-decimal TypeScript libraries for double-entry ledger rules, FIFO lots, portfolio valuation and versioned risk evaluation, with idempotency and reversal invariants.
- Implemented a private Gemini assistant with cited conversation history and pgvector retrieval, with lexical fallback when embeddings are unavailable.

#### BAIOS WORKBENCH | Open-source browser interface

Dec 2025 - Present

- Built and deployed sambai.dev as a 15-app, three-workspace Workbench with multi-instance windows, drag, resize, snap, Atlas navigation and a searchable local archive.
- Designed atomically committed browser-local state, cross-tab revision-conflict handling and schema-validated backup/import without implying remote sync.

### OPEN SOURCE

#### EDGARTOOLS & BETAFLIGHT CONFIGURATOR | Open-source contributions

2026

- Built Blackbox MP4/WebM video export for Betaflight Configurator using WebCodecs, with overlay compositing, progress/ETA, cancellation and streaming saves; PR #5468 is under review for 2026.12.
- Worked through the upstream issue, CI and maintainer review to merge a load-dependent Vitest teardown-race fix for Betaflight Configurator's 2026.12 milestone.
- Repaired legacy SEC 13F TXT parsing so checksum-valid CUSIPs survive column drift; accepted upstream and released in EdgarTools v5.53.0.

### TECHNICAL SKILLS

**Languages & product:** TypeScript, JavaScript, Python, SQL, React, Next.js, Node.js, Vite, Hono, Tailwind CSS

**Data & systems:** PostgreSQL (Neon), MongoDB Atlas, Mongoose, Redis, Drizzle ORM, IndexedDB/Dexie, role/capability authorisation, workers/queues, MCP, vector/lexical retrieval

**Cloud & quality:** Cloudflare Workers, R2, Hyperdrive, Vercel, GitHub Actions, Vitest, Playwright

### EDUCATION

**Bachelor of Applied Information Technology (Software Engineering)** | Wintec

2024